Section 6.7 — Piecewise Functions

Math 103: College Algebra

Building a Piecewise Function

For each list of definitions below, combine the three parts into a single piecewise function. Then, state the domain of the piecewise function.

1. $f(x) = 2x - 4,$ domain $2 < x < 4$
   $f(x) = x^2 - 1,$ domain $4 \leq x < 6$
   $f(x) = x - 2,$ domain $6 \leq x < 8$

2. $g(x) = 5x - 2,$ domain $x > 4$
   $g(x) = 2x - 3,$ domain $x < 2$

3. $g(x) = x,$ domain $x \neq 5$
   $g(x) = -x,$ domain $x = 5$

Evaluating a Piecewise Function

Evaluate each function below for the given values. Write "undefined" if the given value is not in the domain.

4. $f(x) = \begin{cases} -6 & \text{if} \quad x < -3 \\ 3 - x^2 & \text{if} \quad -3 \leq x < 2 \\ -1 & \text{if} \quad x \geq 3 \end{cases}$

   a. $f(-7) =$ b. $f(-3) =$ c. $f(2) =$ d. $f(0) =$

5. $g(x) = \begin{cases} x + 4 & \text{if} \quad x < 4 \\ 2x - 1 & \text{if} \quad x > 7 \end{cases}$

   a. $g(-2) =$ b. $g(12) =$ c. $g(4) =$ d. $g(5) =$

6. $h(x) = \begin{cases} -2 & \text{if} \quad -2.5 < x \leq -1.5 \\ -1 & \text{if} \quad -1.5 < x \leq -0.5 \\ 0 & \text{if} \quad -0.5 < x < 0.5 \\ 1 & \text{if} \quad 0.5 \leq x < 1.5 \\ 2 & \text{if} \quad 1.5 \leq x < 2.5 \end{cases}$

   a. $h(-1.4) =$ b. $h(0.5) =$ c. $h(2) =$ d. $h(2.5) =$

Graphing a Piecewise Linear Function

Graph each function below on a seperate sheet of graph paper or on the other side of this page.

7. $f(x) = \begin{cases} x + 3 & \text{if} \quad x < -1 \\ x - 2 & \text{if} \quad x \geq -1 \end{cases}$

8. $g(x) = \begin{cases} 4 & \text{if} \quad x < -2 \\ 2 - x & \text{if} \quad -2 \leq x \leq 4 \\ -2 & \text{if} \quad x > 4 \end{cases}$

9. $h(x) = \begin{cases} -3 & \text{if} \quad -4 \leq x < -2 \\ -1 & \text{if} \quad -2 \leq x < 0 \\ 1 & \text{if} \quad 0 \leq x < 2 \\ 3 & \text{if} \quad 2 \leq x < 4 \end{cases}$

10. $k(x) = \begin{cases} \frac{2}{3}x - 1 & \text{if} \quad x \neq 3 \\ 4 & \text{if} \quad x = 3 \end{cases}$

11. $f(x) = \begin{cases} (x-1)^2 - 3 & \text{if} \quad x \neq -1 \\ -2 & \text{if} \quad x = -1 \end{cases}$

12. $k(x) = \begin{cases} -2 & \text{if} \quad x < -2 \\ x^2 - 2 & \text{if} \quad -2 \leq x < 1 \\ x + 1 & \text{if} \quad x \geq 1 \end{cases}$